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Absolute Continuity and the Fundamental Theorem of Calculus



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Abstract

The Fundamental Theorem of Calculus [BS11, Theorem 7.3.1] is a theorem with far-reaching consequences across many fields of mathematics, from finding the area between curves, to probabilistic calculus [SB20, p. 7], to solving differential equations. However, the "classical" Fundamental Theorem of Calculus which demands continuity and uses Riemann integration is somewhat limited. The classical Fundamental Theorem of Calculus cannot be applied to an arbitrary function f in cases where the derivative of f is not (Riemann) integrable. It also cannot be applied to functions that are not equal to the indefinite integral of their derivative. This report demonstrates that it is possible to find a form of the Fundamental Theorem of Calculus which holds for a much wider range of functions by replacing uniform continuity with absolute continuity and replacing the Riemann integrals with Lebesgue integrals.

In this report I spent time building up necessary theoretical understandings before discussing absolute continuity and the Fundamental Theorem of Calculus as related definitions. After the main result there is a brief discussion on applications of this theory, particularly in solving classically "ill posed" ordinary differential equations, integration by parts [Fo99] and Opial Inequalities [Ag99; Le99], which themselves have huge applications in differential and difference equations. There are innumerable other applications of this theory which I neglect to discuss in this text, including Lebesgue-Stieltjes integrals, Sobolev spaces, singular functions and more.

In Chapter 1 I discuss the Riemann integral form ("Classical form") of the Fundamental Theorem of Calculus and its shortcomings. In Chapter 2 I explore required concepts such as continuity and functions of bounded variation. Chapter 3 covers the Lebesgue measure and integral. Then in Chapter 4 I define absolute continuity, before proving a series of necessary results around absolute continuity including their structural properties and differentiability. Chapter 5 contains the main result of this report, defining almost everywhere differentiability, the relation between the Lebesgue integral and differentiation before finally examining the Fundamental Theorem of Calculus for absolutely continuous functions. As mentioned before, Chapter 6 contains a discussion on applications of this theory.

The primary references for this project were [RF88], [SS05], [Fo99], [BS11] and [Go94]. I made occasional use of [Bo19], [Th10] [Ji20], [SB20], [Za77] and [Ve10]. I used a number of other resources in aiding my understanding such as [EG92], [JM09], [AB98], [Bu67], [He16] and [Ng20]. For Section 6.3 I primarily used [Ag99] as well as [Le99].

Declaration of Authorship

I, Dylan Sinnott declare that this project "Absolute Continuity and the Fundamental Theorem of Calculus" and all work presented in it is my own, except where otherwise indicated. I further declare that:

- All sources used in this project have been acknowledged and cited.
- No elements of this project have been submitted for a degree at the University of Galway, or at any other University.
- I have acknowledged all assistance received in preparing this project.
- I have read and understood the University of Galway's Academic Integrity Policy, as well as having completed the "Academic Integrity: for Students" certification.

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Chapter 1

Motivation

To understand why absolute continuity is needed for the Fundamental Theorem of Calculus to operate in its most general case, we first must explore the Riemann integral Fundamental Theorem of Calculus. Section 1.1 outlines some basic concepts in calculus before exploring the Riemann integral Fundamental Theorem of Calculus, its proof and necessary assumptions. Section 1.2 explains how we can do better than this form of the fundamental theorem. My primary references for this chapter are [SS05] and [Th10], while I also used [BS11] and [Bo19].

1.1 The (Riemann) Fundamental Theorem of Calculus

The Fundamental Theorem of Calculus is a two part theorem, traditionally known by the "First Form" and "Second Form". The First Form[1] discusses the relationship between differentiation and the indefinite integral of a function. The Second Form swaps the order of differentiation and integration from the First.

First, we need some definitions and theorems to make the Fundamental Theorem meaningful, and to aid in its proof.

Definition 1.1.1 (Derivative)**.** The *derivative* of a function f is given by

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \tag{1.1}$$

For the proof of the first form it is also necessary for me to give the Mean Value Theorem for definite integrals. For this we need a definition of *continuity.*

[1] Across literature there is no consensus on which constitutes the First and Second Forms, the order changes from text to text. In this report I will take the respective forms as per [Th10].

Definition 1.1.2 (Continuity)**.** Let $I \subseteq \mathbb{R}$, $f : I \to \mathbb{R}$ and $u \in I$. f is *continuous* at u if for any $\varepsilon > 0$ $\exists$ $\delta > 0$ such that if x is at any point of I satisfying $|x - u| < \delta \implies |f(x) - f(u)| < \varepsilon$. Otherwise f is discontinuous at u.

An important result relates the derivative and continuity.

Theorem 1.1.3. [Th10, Section 3.2, Theorem 1]

If f has a derivative at $x = c$ then f is continuous at c.

Proof. Given $f'(c)$ exists, to prove f is continuous at c we must show that $\lim_{x\to c} f(x) = f(c)$ or (equivalently) $\lim_{h\to 0} f(c + h) = f(c)$.

If $h \neq 0$ then

$$\begin{aligned} f(c+h) &= f(c) + (f(c+h) - f(c)) \\ &= f(c) + \frac{f(c+h) - f(c)}{h} \cdot h. \end{aligned}$$

Taking the limits as $h \to 0$,

$$\begin{aligned} \lim_{h\to 0} f(c+h) &= \lim_{h\to 0} f(c) + \lim_{h\to 0} \frac{f(c+h) - f(c)}{h} \cdot \lim_{h\to 0} h \\ &= f(c) + f'(c) \cdot 0 \\ &= f(c). \end{aligned}$$

□

Now we can state the Mean Value Theorem for definite integrals.

Theorem 1.1.4 (The Mean Value Theorem for definite integrals)**.** [Th10, Section 5.4, Theorem 3] *If f is continuous on [a,b] then at some point $c \in [a, b]$*

$$f(c) = \frac{1}{b-a} \int_a^b f(x)\ dx.$$

There are a couple of results needed to prove this theorem. Namely the Max-Min Inequality [Th10, Table 5.6] and the Intermediate Value Theorem for Continuous Functions [Th10, Section 2.5].

Proposition 1.1.5 (The Max-Min Inequality)**.** *If $\max f$ and $\min f$ are the minimum and maximum values of f on $[a, b]$, then*

$$\min f \cdot (b-a) \leq \int_a^b f(x)\ dx \leq \max f \cdot (b-a).$$

Proof. Partition the interval $[a, b]$ into $a = x_0 < x_1 < \ ... \ < x_n = b$ such that $\Delta x_n = x_n - x_{n-1}$ and $\sum_{k=1}^{n} \Delta x_k = b - a$. Then for every choice of the points $c_k \in [a, b]$,

$$\begin{aligned}
\min f \cdot (b-a) &= \min f \cdot \sum_{k=1}^{n} \Delta x_k \\
&= \sum_{k=1}^{n} \min f \cdot \Delta x_k \\
&\leq \sum_{k=1}^{n} f(c_k) \Delta x_k \\
&\leq \sum_{k=1}^{n} \max f \cdot \Delta x_k \\
&= \max f \cdot \sum_{k=1}^{n} \Delta x_k \\
&= \max f \cdot (b-a).
\end{aligned}$$

So all Riemann sums for f on $[a, b]$ satisfy

$$\min f \cdot (b-a) \leq \sum_{k=1}^{n} f(c_k) \Delta x_k \leq \max f \cdot (b-a).$$

And so does their limit (the integral).

□

Theorem 1.1.6 (The Location of Roots Theorem)**.** [BS11, Theorem 5.3.5]

If f is a continuous function on $[a, b]$ and $f(a) < 0 < f(b)$ or $f(a) > 0 > f(b)$ then $\exists\ c \in (a, b)$ such that $f(c) = 0$.

Proof. Assume $f(a) < 0 < f(b)$. To prove the theorem we will generate a series of bisections of the interval $[a, b]$. Let $I = [a, b]$ and $I_1 = [a_1, b_1]$ where $a_1 = a, b_1 = b$. Let p_1 be the bisection $p_1 = \frac{1}{2}(a_1 + b_1)$. If $f(p_1) = 0$ then take $c = p_1$ and the theorem is proved.

If $f(p_1) \neq 0$ then either $f(p_1) > 0$ or $f(p_1) < 0$. If $f(p_1) > 0$ then take $a_2 = a_1$ and $b_2 = p_1$ while if $f(p_1) < 0$ take $a_2 = p_1$ and $b_2 = b_1$. Either way, let $I_2 = [a_2, b_2]$ with $I_2 \subset I_1$ and $f(a_2) < 0$, $f(b_2) > 0$.

Continue this bisection process. Assume intervals $I_2, I_3, ..., I_k$ have been created in the same manner. Then $f(a_k) < 0$ and $f(b_k) > 0$ with $p_k = \frac{1}{2}(a_k + b_k)$. As before, if $f(p_k) = 0$ let $c = p_k$ and the theorem is proved. If $f(p_k) > 0$ take $a_{k+1} = a_k$, $b_{k+1} = p_k$ while if $f(p_k) < 0$ take $a_{k+1} = p_k$ and $b_{k+1} = b_k$. Either way let $I_{k+1} = [a_{k+1}, b_{k+1}] \subset I_k$ and $f(a_{k+1}) < 0$. $f(b_{k+1}) > 0$.

If the process terminates at some n (such that $f(p_n) = 0$) then we are done as before. If it doesn't then we have a nested sequence of closed bounded intervals $I_n = [a_n, b_n]$ such that $\forall\ n \in \mathbb{N}$ we have

$$f(a_n) < 0, f(b_n) > 0.$$

But we know that since the intervals have been generated by repeated bisection we have a length $l(I_n) = b_n - a_n = \frac{b-a}{2^{n-1}}$. By the Nested Intervals Property [BS11, Property 2.5.2] $\exists\ c \in I_n\ \forall\ n \in \mathbb{N}$. Since $a_n \leq c \leq b_n\ \forall\ n \in \mathbb{N}$ we have $0 \leq b_n - c \leq b_n - a_n = \frac{b-a}{2^{n-1}}$. So $\lim_{n\to\infty}(a_n) = c = \lim_{n\to\infty}(b_n)$. Since f is continuous at c,

$$\lim_{n\to\infty}(f(a_n)) = f(c) = \lim_{n\to\infty}(f(b_n)).$$

We know that $f(a_n) < 0$ and $f(b_n) > 0\ \forall\ n \in \mathbb{N}$ which implies that $f(c) = \lim_{n\to\infty}(f(a_n)) \leq 0$ and $f(c) = \lim_{n\to\infty}(f(b_n)) \geq 0$. So $f(c) = 0$ and c is a root of f. □

From this theorem comes a more general result, the Intermediate Value Theorem.

Theorem 1.1.7 (The Intermediate Value Theorem for Continuous Functions)**.** *If f is a continuous function on $[a, b]$ and k is any value between $f(a)$ and $f(b)$ $(f(a) < k < f(b))$ then $\exists\ c \in [a, b]$ such that $f(c) = k$.*

Proof. If $a < b$. Let $g(x) = f(x) - k$. By the Location of Roots Theorem

$$\exists\ c \in (a, b) \text{ such that } 0 = g(c) = f(c) - k \implies f(c) = k.$$

If $a > b$ then let $h(x) = k - f(x)$. Therefore

$$\exists\ c \in (b, a) \text{ such that } 0 = h(c) = k - f(c) \implies f(c) = k.$$

□

It is now possible to provide a proof of the Mean Value Theorem.

Proof. [Th10, p. 329] Recall Proposition 1.1.5. Divide both sides of the inequality by $b - a$ to get

$$\min f \leq \frac{1}{b-a}\int_a^b f(x)\ dx \leq \max f.$$

Since f is continuous, by Theorem 1.1.6 f must assume the value $\frac{1}{b-a}\int_a^b f(x)\ dx$ at some point $c \in [a, b]$. □

Now we can state and prove the Fundamental Theorem of Calculus.

Theorem 1.1.8 (The Fundamental Theorem of Calculus (First Form))**.** *If a function f is continuous on $[a, b]$ then $F(x) = \int_a^x f(t)\ dt$ is continuous and differentiable on $[a, b]$ with derivative $f(x)$:*

$$F'(x) = \frac{d}{dx}\int_a^x f(t)\ dt = f(x) \tag{1.2}$$

Proof. [Th10, p. 331]: Apply the definition of the derivative to F(x), for $x, x + h \in (a, b)$:

$$\begin{aligned} F'(x) &= \lim_{h\to 0} \frac{F(x+h) - F(x)}{h} \\ &= \lim_{h\to 0} \frac{1}{h}\left[\int_a^{x+h} f(t)\,dt - \int_a^x f(t)\,dt\right] \\ &= \lim_{h\to 0} \frac{1}{h}\int_x^{x+h} f(t)\,dt. \end{aligned}$$

By the Mean Value Theorem for definite integrals:

$$\frac{1}{h}\int_x^{x+h} f(t)\ dt = f(c)$$

for some $c \in [x, x + h]$.

As $h \to 0, x + h \to x \implies c \to x$ since $c \in [x, x + h]$.

$$\implies \lim_{h\to 0} f(c) = f(x).$$

Putting everything together,

$$\begin{aligned} F'(x) &= \lim_{h\to 0} \frac{1}{h}\int_x^{x+h} f(t)\ dt \\ &= \lim_{h\to 0} f(c) \\ &= f(x). \end{aligned}$$

By choosing $x = a$ or b in Equation (1.1) it then becomes a limit $h \to 0^+$or 0^- respectively. Then f has derivatives over the interval $[a, b]$. By Theorem 1.1.3 f is then continuous on $[a, b]$. □

Theorem 1.1.9 (The Fundamental Theorem of Calculus (Second Form))**.** *If f is continuous over $[a, b]$ and F is any antiderivative of f on $[a, b]$ then*

$$\int_a^b f(x)\ dx = F(b) - F(a) \tag{1.3}$$

Proof. The First Form of the Fundamental Theorem tells us that an antiderivative of f exists

$$G(x) = \int_a^x f(t)\ dt.$$

So if F is *any* antiderivative of f then $F(x) = G(x) + C$ for some constant C. Since both F and G are continuous on $[a, b]$ then

$$\begin{aligned} F(b) - F(a) &= (G(b) + C) - (G(a) + C) \\ &= G(b) - G(a) \\ &= \int_a^b f(t)\ dt\ - \int_a^a f(t)\ dt \\ &= \int_a^b f(t)\ dt. \end{aligned}$$

□

If the indefinite integral of f cannot be found, then we cannot use the Fundamental Theorem of Calculus under our list of current assumptions.

1.2 Why the (Riemann) Fundamental Theorem of Calculus is insufficient

We've already seen a problem with the Fundamental Theorem of Calculus, in particular its Second Form. Differentiability of a function does not guarantee an integrable derivative. So to assume that Equation (1.3) holds for all continuous functions is in fact dangerous; there are continuous functions F that are nowhere differentiable, or that have non-integrable $F'(x)$. For the Fundamental Theorem to be consistent, it is essential to find what conditions on $F(x),\ x \in [a,b]$ to guarantee $F'(x)$ exists (for a.e[2] x).

a.e differentiability is an important concept for refining the Fundamental Theorem of Calculus, which that is covered in depth in Section 5.1. For now we only need consider that there are some functions that are not equal to the indefinite integral of their derivative and so the Fundamental Theorem of Calculus cannot be applied. We will now see an example of such a function.

Example 1.2.1 (The Cantor Ternary Function)**.** For $x \in [0,1]$ $\exists$ an infinite series $a_1, a_2, ...$ where each $a_n \in \{0,1,2\}$ such that

$$x = \sum_{n=1}^{\infty} \frac{a_n}{3^n}$$

This is the ternary representation of x.

Let $N(x)$ be the first n such that $a_n = 1$ in the ternary representation. If x has multiple ternary representations then choose $N(x)$ as the largest possible value of all ternary representations. If x has a ternary representation with no $a_n = 1 \implies N(x) = \infty$.

The Cantor set is the set consisting of all $x \in [0,1]$ for which $N(x) = \infty$. This can be written as all x of the form

$$x = \sum_{n=1}^{\infty} \frac{2b_n}{3^n}$$

where $b_n \in \{0,1\}, b_n = 0$ if $a_n = 0$ and $b_n = 1$ if $a_n = 2$.

Equivalently, we can say that the Cantor set is constructed using the following algorithm.

1. Begin with the interval $[0,1]$.
2. Divide [0.1] into three subintervals and remove the open middle subinterval $(\frac{1}{3}, \frac{2}{3})$.
3. Divide the two remaining subintervals into three subintervals and remove the open middle subintervals $(\frac{1}{9}, \frac{2}{9})$ and $(\frac{7}{9}, \frac{8}{9})$.
4. Continue this process indefinitely.

More precisely,

[2]For almost every x. A precise definition will come later.

Definition 1.2.2. Let $x \in [0,1]$. Taking x in its ternary representation $x = \sum_{n=1}^{\infty} \frac{a_n}{3^n}$ with $a_n \in \{0,1,2\}$. As before $b_n = 0$ if $a_n = 0$ and $b_n = 1$ if $a_n = 2$. The Cantor Ternary Function is then

$$C(x) = \sum_{n=1}^{\infty} \frac{b_n}{2^n} \tag{1.4}$$

The Cantor ternary function can be interpreted as a function that maps the Cantor set onto the interval $[0,1]$. It can be shown that the Cantor ternary function is *continuous*.

Proposition 1.2.3. *The Cantor Ternary Function is continuous.*

Proof. [DM06]

From Equation (1.4) the Cantor ternary function is increasing and constant on each open interval. We can take the ternary representations of $x, y \in [0,1], x \neq y$ as $x = \sum_{m=1}^{\infty} \frac{a_m}{3^m}$ and $y = \sum_{m=1}^{\infty} \frac{a'_m}{3^m}$. Let n be the first index such that $a_n \neq a'_n$. Then $|x-y| \geq \frac{1}{3^n}$, $|C(x)-C(y)| \leq \frac{1}{2^n}$.

Given $\varepsilon > 0$, choose $k \in \mathbb{N}$ such that $\frac{1}{2^k} < \varepsilon$, and set $\delta = \frac{1}{3^k}$. If $|x-y| < \delta$, then $n > k$ and hence

$$|C(x) - C(y)| \leq \frac{1}{2^n} < \frac{1}{2^k} < \varepsilon.$$

So the Cantor ternary function is continuous. □

Example 1.2.4. On every interval that is removed when constructing the Cantor set, the Cantor ternary function is constant. For example

- On $(\frac{1}{3}, \frac{2}{3}) : C(x) = \frac{1}{2}$
- On $(\frac{1}{9}, \frac{2}{9}) : C(x) = \frac{1}{4}$
- On $(\frac{7}{9}, \frac{8}{9}) : C(x) = \frac{3}{4}$

Note that these intervals are exactly those that were removed in the first three steps of constructing the Cantor set above.

Graphically, the Cantor ternary function appears as a staircase climbing gradually from $0 \to 1$, with the "steps" being the horizontal sections.

An approximate illustration of this function's evolution is seen in Figure 1.1.

Of course this is not an accurate demonstration of the Cantor function's nature, it is made up of a series of open intervals where over each individual interval $C(x)$ is constant, while the graph shows a continuous function that increases up to $C(x) = \frac{1}{2}$ before then continuing to

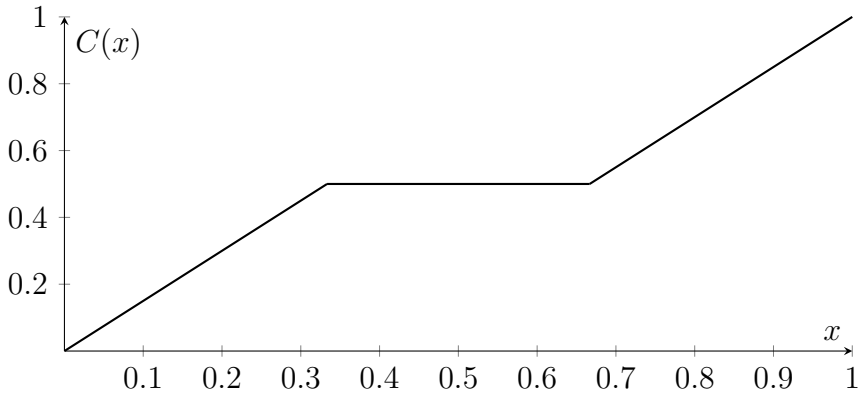


Figure 1.1: Approximate evolution of the Cantor ternary function

1. In reality the increasing segments represent the "steps" that earn the Cantor function the nickname "Devil's Staircase."

C is constant on each individual open interval. So C is differentiable on each interval with $C' = 0 \implies C$ is differentiable a.e. and $C' = 0$ where it exists.

But

$$C(1) - C(0) = 1 \text{ and } \int_0^1 C'(x)\, dx = 0$$

.

$$1 \neq 0$$

So we can say that the Cantor ternary function is not the indefinite integral of its derivative and the (Riemann) Fundamental Theorem of Calculus is not valid for it. Another important fact to note, the Cantor ternary function is uniformly continuous. Because it is continuous on a closed and bounded interval it is uniformly continuous (see Definition 2.1.1). This is important: we have found a uniformly continuous function for which the (Riemann) Fundamental Theorem of Calculus does not apply. So, continuity and uniform continuity are too weak definitions for the Fundamental Theorem to hold in general.

Chapter 2

Continuity and functions of bounded variation

Before exploring absolute continuity and going on to consider the Fundamental Theorem of Calculus for absolutely continuous functions, there are several background concepts that must be understood. Section 2.1 defines uniform continuity and Lipschitz continuity, as well as exploring their properties and why neither are the optimal class of functions to consider the Fundamental Theorem of Calculus. Section 2.2 defines variation, functions of bounded variation as well as a brief look at monotone functions and revisiting the the Cantor ternary function. My primary references for this chapter are [RF88], [SS05] and [BS11], while I also made use of [Go94], [EG92] and [Ji20].

2.1 Uniform continuity and stronger notions

We defined before the concept of continuity. A related definition is that of *uniform continuity*.

Definition 2.1.1 (Uniform continuity)**.** Let $I \subseteq \mathbb{R}$ and $f : I \to \mathbb{R}$. f is *uniformly continuous* on I if for each $\varepsilon > 0$ $\exists$ δ such that if $x, u \in I$ satisfy $|x - u| < \delta \implies |f(x) - f(u)| < \varepsilon$.

Remark. A function that is *uniformly continuous* on I is continuous at every point of I, but the converse is not generally true.

This is a more stringent definition than that of simple continuity as it governs the behaviour of the function for all points on I rather than at a single point. An even stronger definition is *Lipschitz*.

Definition 2.1.2 (Lipschitz)**.** Let $I \subseteq \mathbb{R}$ and let $f : I \to \mathbb{R}$. If $\exists$ a constant $L > 0$ such that $|f(x) - f(u)| \leq L|x - u|$ $\forall$ $x, u \in I$ then f is *Lipschitz* on I.

The previous two definitions carry the following result.

Theorem 2.1.3. [BS11, Theorem 5.4.5] *All Lipschitz functions are uniformly continuous.*

Proof. If f is Lipschitz then by Definition 2.1.2:

$$|f(x) - f(u)| \leq L|x - u|.$$

Given $\varepsilon > 0$ take $\delta = \frac{\varepsilon}{L}$. If $x, u \in I$ satisfy $|x - u| < \delta$, then

$$|f(x) - f(u)| < L \cdot \frac{\varepsilon}{L} = \varepsilon.$$

$\implies$ f is uniformly continuous on I.

□

As seen in Example 1.2.1 the Cantor ternary function is uniformly continuous, and serves as an example of a uniformly continuous function for which the Fundamental Theorem of Calculus does not apply, showing uniform continuity to be too weak of a condition, including functions for which the Fundamental Theorem of Calculus does not apply.

It turns out that Lipschitz is not the right definition either, it excludes functions for which the Fundamental Theorem of Calculus is valid.

Example 2.1.4. $f(x) = \sqrt{x}$, $x \in [0, 1]$ is not Lipschitz but the Fundamental Theorem of Calculus holds perfectly. We will show f is not Lipschitz by contradiction. Assume $\exists\ L > 0$ such that

$$|\sqrt{x} - \sqrt{y}\ | \leq L|x - y| \ \forall\ x, y \in [0, 1].$$

Take $y = 0$. So $\forall\ x \in (0, 1]$:

$$\sqrt{x} \leq Lx \implies \frac{1}{\sqrt{x}} \leq L$$

As $x \to 0^+$, $\frac{1}{\sqrt{x}} \to \infty$. So no L can be found such that f is Lipschitz $\implies$ $f(x)$ is not Lipschitz.

Applying the Fundamental Theorem of Calculus (Second Form): $\int_a^b f'(x)\ dx = f(b) - f(a)$

$$f(x) = \sqrt{x} \implies f'(x) = \frac{1}{2\sqrt{x}}$$

Note, $f(1) - f(0) = 1$ and $\int_0^1 \frac{1}{2\sqrt{x}}\ dx = 1 = f(1) - f(0)$

So $f(x) = \sqrt{x}$ is not Lipschitz but the Fundamental Theorem of Calculus holds.

We have seen now that there are some functions that are not Lipschitz but for which the Fundamental Theorem of Calculus holds. We have also seen that there are some uniformly continuous functions for which the Fundamental Theorem of Calculus doesn't hold.

Neither of these classes of functions are the ideal choice of a class of functions to which we can apply the Fundamental theorem of Calculus. We will see that the ideal class is that of absolutely continuous functions.

2.2 Functions of bounded variation

It is necessary when thinking about absolutely continuous functions to first have an understanding of functions of bounded variation. First consider bounded functions.

Definition 2.2.1 (Bounded functions)**.** A function $f : I \to \mathbb{R}$ is bounded if on I if $\exists$ a constant $M > 0$ such that $|f(x)| \leq M \ \forall \ x \in I$.

Notably this is *not* the same as functions of bounded variation. All bounded variation functions are bounded but the converse is not necessarily true.

Definition 2.2.2 (Variation)**.** The variation of f on $[a, b]$ is:

$$V(f, [a, b]) = \sup\left\{ \sum_{k=1}^{n} |f(b_k) - f(a_k)| \right\}$$

where the suprenum is over all finite collections $\{[a_k, b_k] : k \in [1, n]\}$ of non-overlapping intervals in $[a, b]$. [Go94, p. 49]

Definition 2.2.3 (Bounded Interval)**.** An interval I is bounded if $\forall \ x \in I \ \exists \ b$ such that $x \leq b$ and $\exists \ a$ such that $x \geq a$ (I has both an upper and lower bound).

With this understanding of variation and bounded intervals, bounded variation follows:

Definition 2.2.4 (Bounded Variation)**.** f is of bounded variation on $[a, b]$ if the variation $V(f, [a, b])$ is finite.

An equivalent definition is:

If $\exists \ M < \infty$ such that $\sum_{i=1}^{n} |f(t_i) - f(t_{i-1})| \leq M \ \forall$ partitions then f is of bounded variation. The space of all such f is denoted $BV([a, b])$.

Functions of bounded variation have the following properties:

1. If f is of bounded variation on $[a, b]$ then f is differentiable a.e on $[a, b]$. [SS05, Theorem 3.4, p. 121]. (This also follows from Theorem 2.2.8 and Proposition 3.1.15).
2. If f is of bounded variation on $[a, b]$ then f is bounded on $[a, b]$.
3. If f is of bounded variation on $[a, b]$ then the set of points in $[a, b]$ at which f is discontinuous is countable. [Fo99, Theorem 3.27]

4. If f is differentiable at every point of $[a, b]$ and f' is bounded then f is of bounded variation on $[a, b]$.

5. If f is of bounded variation on $[a, b]$ then f', where it exists is also measurable on $[a, b]$. [Ng20, Theorem 6], [SS05, Corollary 3.7].

I will present the proof for 2 and 4 above.

Proof. Starting with property 2. Since f is of bounded variation $\exists\ M$ such that $\sum_{i=1}^{n} |f(t_i) - f(t_{i-1})| \leq M\ \forall$ partitions $[t_{i-1}, t_i]$ of the interval $[a, b]$.

Let $x \in [a, b]$. Consider the partition consisting of just the two points x, a. Since f is of bounded variation on $[a, b]$ we have $|f(x) - f(a)| \leq M$.

Applying the triangle inequality gives,

$$|f(x)| \leq |f(a)| + |f(x) - f(a)| \leq |f(a)| + M.$$

But $|f(a)| + M$ is a finite constant, so $f(x)$ is bounded on $[a, b]$.

For property 4, assume f is differentiable on $[a, b]$ with f' bounded. We have,

$$|f'(x) \leq M| \ \forall\ x \in [a, b]$$

Partition $[a, b]$ by taking $a = t_0 < t_1 < \ \ldots\ < t_N = b$. Since f is continuous and differentiable on $[a, b]$ we can apply the Mean Value Theorem, for each partition $[t_{i-1}, t_i]\ \exists\ c_i$ such that

$$f(t_i) - f(t_{i-1}) = f'(c_i)(t_i - t_{i-1})$$

Taking the absolute value and using the bound on f',

$$|f(t_i) - f(t_{i-1})| = |f'(c_i)||t_i - t_{i-1}| \leq M(t_i - t_{i-1})$$

since $t_i > t_{i-1}$. Summing over all partitions gives,

$$\sum_{i=1}^{n} |f(t_i) - f(t_{i-1})| \leq M \sum_{i=1}^{n} (t_i - t_{i-1}) = M(b - a).$$

So $V(f, [a, b]) < \infty$ and f is of bounded variation on $[a, b]$.

□

The variation of a function of bounded variation can be classified as follows.

Lemma 2.2.5. [Go94, Chapter 4, Lemma 4.2] *If $f : [a, b] \to \mathbb{R}$ is of bounded variation on $[a, b]$ then*

1. *If $f : [a, b] \to \mathbb{R}$ is of bounded variation on $[a, b]$ then f is of bounded variation on every subinterval of $[a, b]$ and*
$$V(f, [a, b]) = V(f, [a, c]) + V(f, [c, b]) \ \forall \ c \in [a, b]$$
2. *If $f : [a, b] \to \mathbb{R}$ is of bounded variation on $[a, c]$ and $[c, d]$ then f is of bounded variation on $[a, b]$.*

Proof. If f is of bounded variation on $[a, b]$ then clearly $V(f, [c, d]) \leq V(f, [a, b])$ for each $[c, d] \subseteq [a, b]$. Hence f is of bounded variation on every subinterval of $[a, b]$. Let $c \in (a, b)$ and $\{[a_k, b_k] : 1 \leq k \leq n\}$ be any collection of disjoint intervals in $[a, b]$. By (if necessary) splitting the interval we may assume that either $[a_k, b_k] \subseteq [a, c]$ or $[a_k, b_k] \subseteq [c, b]$ for each k. Then

$$\begin{aligned}\sum_{k=1}^{n} |f(b_k) - f(a_k)| &= \sum_{b_k \leq c} |f(b_k) - f(a_k)| + \sum_{a_k \geq c} |f(b_k) - f(a_k)| \\ &\leq V(f, [a, c]) + V(f, [c, b])\end{aligned}$$

$$\implies V(f, [a, b]) \leq V(f, [a, c]) + V(f, [c, b]).$$

The second property is now proved. To complete the proof of the first let $\varepsilon > 0$ and choose disjoint intervals $\{[s_i, t_i] : 1 \leq i \leq m\}$ in $[a, c]$ and $\{[u_j, v_j] : 1 \leq j \leq n\}$ in $[c, b]$ such that

$$\sum_{i=1}^{m} |f(t_i) - f(s_i)| > V(F, [a, c]) - \frac{\varepsilon}{2}$$

$$\sum_{j=1}^{n} |f(v_j) - f(u_j)| > V(F, [c, b]) - \frac{\varepsilon}{2}.$$

Then

$$V(f, [a, b]) \geq \sum_{i=1}^{m} |f(t_i) - f(s_i)| + \sum_{j=1}^{n} |f(v_j) - f(u_j)|$$

$$> V(f, [a, c]) + V(f, [c, b)] - \varepsilon$$

$$\geq V(f, [a, c]) + V(f, [c, b)]$$

since $\varepsilon > 0$ was arbitrary. □

Example 2.2.6 (The Cantory ternary function (revisited))**.** It can be shown that the Cantor ternary function (Example 1.2.1) is of bounded variation. The Cantor function is a continuous monotone increasing function on $[a, b] \implies$ it is of bounded variation [BS11, Section 5.6].

Definition 2.2.7 (Monotone Increasing function)**.** A monotone function defined on an interval I is a function that is either strictly increasing or strictly decreasing on I. Furthermore, a *monotone increasing function* is a monotone function that is strictly increasing. A function $f : I \to \mathbb{R}$ is *monotone increasing function* if $\forall\ x_i, x_{i+1} \in I$ with $x_i \leq x_{i+1}$: $f(x_i) \leq f(x_{i+1})$.

Monotone increasing functions have some interesting properties that directly relate to functions of bounded variation.

Theorem 2.2.8 (Jordan decomposition)**.** [RF88, Chapter 5, Theorem 5] *If a function f is of bounded variation on $[a, b]$ then it can be split into two monotone increasing functions*

$$f = f^+ - f^-.$$

Proof. For each $x \in [a, b]$ the total variation of f on $[a, x]$ is defined as in Definition 2.2.2. Let

$$V(x) = V(f, [a, x]).$$

Since f is of bounded variation on $[a, b]$, $V(x)$ is finite $\forall\ x$ and it is monotone increasing.

Then, $\forall\ x, y \in [a, b]$ since $V(y) - V(x)$ encodes all total variation in the function, while $|f(y) - f(x)|$ gives only one contribution to variation we have,

$$V(y) - V(x) \geq |f(y) - f(x)|.$$

Take f^+ and f^- to be two functions such that $f = f^+ - f^-$. Specifically we will use the fact that $V(x)$ is monotone increasing and use $V(y) - V(x) \geq |f(y) - f(x)|$. Let

$$f^+(x) = \frac{1}{2}\big(f(x) + V(x)\big)$$

and

$$f^-(x) = \frac{1}{2}\big(V(x) - f(x)\big).$$

Then

$$f^+(x) - f^-(x) = \frac{1}{2}\big(f(x) + V(x)\big) - \frac{1}{2}\big(V(x) - f(x)\big) = f(x).$$

Again, take $x, y \in [a, b]$. Then

$$f^+(y) - f^+(x) = \frac{1}{2}\big(f(y) - f(x) + V(y) - V(x)\big).$$

But $V(y) - V(x) \geq |f(y) - f(x)| \geq f(y) - f(x) \implies f^+$ is monotone increasing.

Similarly,

$$f^-(y) - f^-(x) = \frac{1}{2}\big(V(y) - V(x) - (f(y) - f(x))\big),$$

So f^- is also monotone increasing as above.

$\implies f$ can be decomposed into two monotone increasing functions.

□

We will see in Proposition 3.1.15 an important result that applies to monotone increasing functions, they are differentiable almost everywhere. To prove it we will need to make use of an important result that follows in Chapter 3, namely the Vitali Covering Lemma, as well as the following definition of upper and lower derivatives.

Definition 2.2.9 (Upper and lower derivatives)**.** Let $f : [a, b] \to R$. The upper and lower right derivatives of f at $x \in [a, b)$ are

$$D^+f(x) = \lim_{\delta \to 0^+} \sup\{\frac{f(y) - f(x)}{y - x} : x < y < x + \delta\}$$

$$D_+f(x) = \lim_{\delta \to 0^+} \inf\{\frac{f(y) - f(x)}{y - x} : x < y < x + \delta\}$$

Similarly, the upper and lower left derivatives of f at $(a, b]$ are

$$D^-f(x) = \lim_{\delta \to 0^+} \sup\{\frac{f(y) - f(x)}{y - x} : x - \delta < y < x\}$$

$$D_-f(x) = \lim_{\delta \to 0^+} \inf\{\frac{f(y) - f(x)}{y - x} : x - \delta < y < x\}$$

Remark. A function f is differentiable at $x \in (a, b)$ if al four derivatives are finite and equal. [Go94, p. 54].

Chapter 3

Lebesgue Theory

It is necessary at this point to include a discussion on Lebesgue and measure theory, with the goal of being able to define differentiability a.e (which we will do in Section 5.1). Lebesgue integration is particularly important as through it we can by bypass the rigidity of Riemann integration.

In Section 3.1 and Section 3.2 I will explore the concept of measure, null sets, an introduction to what almost everywhere means, the Lebesgue integral and a comparison between Lebesgue and the Riemann integration. My primary references for this chapter are [RF88], [SS05] and [BS11], while I also made use of [Go94], [EG92] and [Ji20].

3.1 Lebesgue measure

Definition 3.1.1 (Null sets)**.** A set $Z \subset \mathbb{R}$ is a *null set* if $\forall\ \varepsilon > 0\ \exists$ a countable collection $\{(a_k, b_k)\}_{k=1}^{\infty}$ of open intervals such that

$$Z \subset \bigcup_{k=1}^{\infty}(a_k, b_k) \text{ and } \sum_{k=1}^{\infty}(b_k - a_k) \leq \varepsilon$$

Example 3.1.2 (Some examples of null sets)**.**

1. $\mathbb{Q}$, the set of rational numbers.
2. $\mathbb{N}$, the set of natural numbers.
3. The Cantor set (uncountable but still a null set).

Definition 3.1.3 (Almost everywhere)**.** If $Q(x)$ is a statement about the point $x \in I$ then $Q(x)$ holds *almost everywhere* on I if $\exists$ a null set $Z \subset I$ such that $Q(x)$ holds $\forall\ x \in I \setminus Z$.

$$\text{i.e, we say } Q(x) \text{ holds for a.e. } x \in I$$

Another way to define a null set is to say that it is a set of *measure* 0.

Definition 3.1.4 ((Lebesgue[1]) Outer Measure)**.** The *(Lebesgue) outer measure* $m^*(Z)$ of a set Z is the infimum over all countable coverings of Z by open intervals: $Z \subset \bigcup I_n$. So the outer measure of Z is given by

$$m^*(Z) = \inf_{Z \subset \bigcup_{I_n}} \sum l(I_n)$$

Where the length of an interval $l(I_n)$ is given by the difference in its endpoints[2].

The outer measure holds some useful properties. We will establish 3 propositions.

Proposition 3.1.5. [RF88, Section 3.2, Proposition 1]

The outer measure of an interval is its length.

Proof. Consider a closed finite interval $[a, b]$. The open interval $(a-\varepsilon, b+\varepsilon)$ contains $[a, b]$ $\forall\ \varepsilon > 0$.

$$\implies m^*[a, b] = \leq l(a - \varepsilon, b + \varepsilon) = b - a + 2\varepsilon$$

Since $m^*[a, b] \leq b - a + 2\varepsilon\ \forall\ \varepsilon > 0$ we have

$$m^*[a, b] \leq b - a$$

.

We now must show that $m^*[a, b] \geq b - a$. Equivalently, we must show that if $\{I_n\}$ is any collection of finite open intervals covering $[a, b]$ then

$$\sum l(I_n) \geq b - a \tag{3.1}$$

By the Heine Borel theorem [RF88, p. 56], if a collection of open intervals covers $[a, b]$ then it contains a finite subcollection that also covers $[a, b]$. Since the sum of lengths of the finite subcollection is no more than the sum of lengths of the original collection of open intervals (3.1) is proved for finite collections $\{I_n\}$ that cover $[a, b]$.

Since $a \in \bigcup I_n$ a must be contained in one of the open intervals. Denote this open interval $(a_1, b_1) \implies a_1 < a < b_1$. If $b_1 \leq b \implies b_1 \in [a, b]$ and since $b_1 \notin (a_1, b_1)$ there must be another open interval (a_2, b_2) in $[a, b]$ such that $b_1 \in (a_2, b_2) \implies a_2 < b_1 < b_2$.

Continuing in this fashion we get a sequence $(a_1, b_1), ..., (a_k, b_k)$ from the collection of I_n's such that $a_i < b_{i-1} < b_i$. We have a *finite* collection of intervals so our process terminates when we find an interval such that $b \in (a_k, b_k) \implies a_k < b < b_k$. Thus

[1]While there are other outer measures, these will not be considered in this text. So within this chapter I will use the terms Lebesgue outer measure and outer measure interchangeably.

[2]This is demonstrated in the proof of Proposition 3.1.5

$$\begin{aligned}\sum l(I_n) &\geq \sum l(a_i, b_i) \\ &= (b_k - a_k) + (b_{k-1} - a_{k-1}) \ + \ ... \ + \ (b_1 - a_1) \\ &= b_k - (a_k - b_{k-1}) - (a_{k-1} - b_{k-2}) \ - \ ... \ - (a_2 - b_1) - a_1 \\ &> b_k - a_1\end{aligned}$$

since $a_i < b_{i-1}$. But $b_k > b$ and $a_1 < a \implies b_k - a_1 > b - a$

$$\implies m^*[a, b] = b - a$$

.

To demonstrate that the length of an interval is in fact the difference in its endpoints:

Consider a finite interval I of which the closed interval $J \subset I$. If $\varepsilon > 0 \implies l(J) > l(I) - \varepsilon$. Hence

$$l(J) = l(J) = m^*(J) \leq m^*(I) = l(I).$$

So for each $\varepsilon > 0$,

$$l(I) - \varepsilon \leq m^*(I) \leq l(I).$$

So $m^*(I) = l(I)$ for any finite interval I.

If I is infinite then for any real number c there is a closed interval $J \subset I$ with $l(J) = c$. Hence $m^*(I) \geq m^*(J) = l(J) = c$. Since $m^*I \geq c$ for each c then

$$m^*(I) = \infty = l(I).$$

□

Proposition 3.1.6. [RF88, Section 3.2, Proposition 2] *For a countable collection of sets of real numbers* $\{A_n\}$,

$$m^*(\bigcup A_n) \leq \sum m^*(A_n)$$

Proof. If any A_n has infinite measure then clearly the inequality holds. If $m^*(A_n)$ is finite then given $\varepsilon > 0$ $\exists$ a countable collection of open intervals $\{I_{n,i}\}_i$ such that $A_n \subset \bigcup_i \{I_{n,i}\}_i$ and $\sum_i l(I_{n,i}) < m^*(A_n) + 2^{-n}\varepsilon$. Now the collection of open intervals $\{I_{n,i}\}_i$ is countable and covers $\bigcup A_n$. Thus,

$$m^*(\bigcup A_n) \leq \sum_{n,i} l(I_{n,i}) = \sum_n \sum_i l(I_{n,i}) < \sum_n (m^*(A_n) + 2^{-n}\varepsilon) = \sum m^*(A_n) + \varepsilon$$

Since $\varepsilon > 0$ is arbitrary,

$$m^*(\bigcup A_n) \leq \sum m^*(A_n).$$

□

Corollary 3.1.7. *For all countable sets A we have $m^*(A) = 0$.*

Corollary 3.1.8. $m^*(A + h) = m^*(A)$ $\forall$ $h \in \mathbb{R}$.

Definition 3.1.9 (Measurable sets)**.** A set A is (Lebesgue) *measurable* if $\forall$ $\varepsilon > 0$ $\exists$ an open set O with $A \subset O$ and $m^*(O - A) \leq \varepsilon$.

Definition 3.1.10 (Open sets)**.** A set A is *open* if $\forall$ $x \in A$, A includes an interval centered at x. That is, $\exists$ $\delta > 0$ such that $(x - \delta, x + \delta) \subset A$.

We can now define the notion of *measurable functions*.

Definition 3.1.11 (Measurable functions)**.** A function f defined on a measurable subset $A \subset \mathbb{R}^d$ is *measurable* if $\forall$ $a \in \mathbb{R}$ the set $f^{-1}([-\infty, a]) = \{x \in A : f(x) > a\}$ is measurable.

Remark. If f is continuous on $\mathbb{R}^d$, then f is measurable.

Measurable functions are important as a function must be measurable to be Lebesgue integrable.[3]

There is an important result related to measurable functions and functions of bounded variation.

Proposition 3.1.12. *If f is of bounded variation on $[a, b]$ then f is measurable on $[a, b]$.*

[3]Note this is not the same as saying if a function is measurable it is Lebesgue integrable, which is not generally true.

Proof. Use the Jordan Decomposition theorem Theorem 2.2.8, so $f = f^+ - f^-$ where f^+ and f^- are monotone increasing. f^+ is measurable if the set $(f^+)^-([a,b]) = \{x \in [a,b] : f^+ > \alpha\}$ is measurable. Since f^+ is monotone increasing this either takes the form of an empty interval, the whole interval $[a,b]$ or some subinterval $[c,b]$ or $(c,b]$ for some $c \in [a,b]$. In all cases intervals are Lebesgue measurable so f is measurable. □

The Lebesgue measure allows us to define an important result which we will use in the proof of Proposition 3.1.15, that of Vitali covers and the Vitali Covering Lemma, which we will state without proof.[4]

Definition 3.1.13 (Vitali cover)**.** For a subset $A \subseteq \mathbb{R}$, a collection I^* of intervals is a *Vitali cover* of A if $\forall\ x \in A$ and $\varepsilon > 0$ $\exists$ an interval $I \in I^*$ such that $x \in I$ and the measure $(m^*(I))$ of $I < \varepsilon$.

Lemma 3.1.14 (The Vitali Covering Lemma)**.** *Let $A \subseteq \mathbb{R}$ where A has finite measure. If I^* is a Vitali cover of A then $\forall\ \varepsilon > 0$ $\exists$ a finite collection $\{I_k : 1 \leq k \leq n\}$ of disjoint intervals in I^* such that*

$$m^*(A - \bigcup_{k=1}^{n} I_k) < \varepsilon$$

In addition, $\exists$ a sequence $\{I_k\}$ of disjoint intervals in I^* such that

$$m^*(A - \bigcup_{k=1}^{\infty} I_k) = 0$$

Proposition 3.1.15. [Go94, p. 55] *Monotone increasing functions are differentiable almost everywhere.*

Proof. First we will show that for monotone increasing functions on $[a,b]$ all four derivatives are finite almost everywhere on $[a,b]$. Assume f is monotone increasing on $[a,b]$.

It is implicit that the derivatives are non-negative. To show that the upper right derivative is finite almost everywhere on $[a,b]$, assume $m^*(A) = \alpha > 0$ where $A = \{x \in [a,b) : D^+f(x) = \infty\}$. Choose $M > 0$ such that $\frac{M\alpha}{2} > f(b) - f(a)$. $\forall\ x \in A$ $\exists$ a decreasing sequence $\{y_n^x\}$ that converges to x such that

$$\frac{f(y_n^x) - f(x)}{y_n^x - x} \geq M\ \forall\ n$$

The collection $\{[x, y_n^x] : x \in A, n \in \mathbb{Z}^+\}$ is a Vitali cover of A. By the Vitali Covering Lemma $\exists$ a disjoint collection $\{[x_i, y_i] : 1 \leq i \leq n\}$ of these intervals such that $\sum_{i=1}^{n}(y_i - x_i) > \frac{\alpha}{2}$. But then

$$\sum_{i=1}^{n}(f(y_i) - f(x_i)) \geq \sum_{i=1}^{n} M(y_i - x_i) > \frac{M\alpha}{2} > f(b) - f(a)$$

[4]For a proof of the Vitali Covering Lemma, see [Go94, p. 52]

which is a contradiction. So $m^*(A) = 0$. A similar process can be repeated for the upper left derivative to demonstrate that all four derivatives are finite almost everywhere on $[a, b]$. Now we need only show that the four derivatives equal almost everywhere. We will show that the set $B = \{x \in (a, b) : D_+f(x) < D^+f(x)\}$ has measure zero. Suppose $m^*(B) > 0$. Then $\exists\ p, q \in \mathbb{Q}$ such that

$$C = \{x \in B : D_+f(x) < p < q < D^+f(x)\}$$

satisfies $m^*(C) = \gamma > 0$. Let $\varepsilon > 0$. Then, $\exists$ an open set $O \subseteq (a, b)$ such that $C \subseteq O$ and $m^*O < \gamma + \varepsilon$. For each $x \in C$ and $n \in \mathbb{Z}^+$ $\exists\ y_n^x$ such that $x < y_n^x < x + \frac{1}{n}, [x, y_n^x] \subseteq O$ and

$$\frac{f(y_n^x) - f(x)}{y_n^x - x} < p$$

$\{[x, y_n^x] : x \in C, n \in \mathbb{Z}^+$ is then a Vitali cover of C. So $\exists$ a disjoint collection $\{[x_i, y_i] : 1 \leq i \leq m\}$ of these intervals such that $m^*(C - \bigcup_{i=1}^m [x_i, y_i]) < \varepsilon$. So

$$\sum_{i=1}^{m}(f(y_i) - f(x_i)) < p\sum_{i=1}^{m}(y_i - x_i) \leq pm^*O < p(\gamma + \varepsilon)$$

Take $D = C \cap (\bigcup_{i=1}^m (x_i, y_i))$ so $m^*(C) > \gamma - \varepsilon$. $\forall\ u \in D, n \in \mathbb{Z}^+$ $\exists\ v_n^u$ such that $u < v_n^u < u + \frac{1}{n}, [u, v_n^u] \subseteq (x_i, y_i)$ for some i. Then

$$\frac{f(v_n^u) - f(u)}{v_n^u - u} > q$$

So $\{[u, v_n^u] : u \in D, n \in \mathbb{Z}^+\}$ is a Vitali cover of D. Again by the Vitali Covering Lemma $\exists$ a disjoint collection of intervals $\{[u_j, v_j] : 1 \leq j \leq n\}$ such that $\sum_{j=1}^n (v_j - u_j) > m^*D - \varepsilon$. Here

$$\sum_{j=1}^{n}(f(v_j) - f(u_j)) > q\sum_{j=1}^{n}(v_j - u_j) > q(m^*D - \varepsilon) > q(\gamma - 2\varepsilon)$$

$\forall\ i \in [1, m]$ let $\pi_i = \{[u_j, v_j] \subseteq (x_i, y_i)\}$. Since f is monotone increasing

$$\begin{aligned} q(\gamma - 2\varepsilon) &< \sum_{j=1}^{n}(f(v_j) - f(u_j)) = \sum_{i=1}^{m}\sum_{j\in\pi_i}(f(v_j) - f(u_j) \\ &\leq \sum_{i=1}^{n}(f(y_i) - f(x_i)) \\ &< p(\gamma + \varepsilon) \end{aligned}$$

Since ε was chosen arbitrarily we have $q\gamma \leq p\gamma$ which is a contradiction. So the set B must have measure zero and hence, f must be differentiable almost everywhere.

□

3.2 Lebesgue Integration

We have seen before the Riemann integral. In order to refine the Fundamental Theorem of Calculus (in essence, to consider it for absolutely continuous functions) its Riemann integrals must be interchanged with Lebesgue integrals.

We will build up an understanding of Lebesgue integration by first considering the Lebesgue integral of characteristic and simple functions before moving towards general Lebesgue integrable functions.

Up until this point we have used $\int$ to represent the Riemann integral. From this point onwards we will use $\int$ to represent the Lebesgue integral unless otherwise stated. The Riemann integral will be represented by $R\int$.

Definition 3.2.1 (Characteristic functions)**.** A function $\chi_A(x)$ is a characteristic function of a measurable set A if

$$\chi_A(x) = \begin{cases} 1, x \in A \\ 0, x \notin A \end{cases} \tag{3.2}$$

Definition 3.2.2 (The Lebesgue Integral of a Characteristic Function)**.** The Lebesgue integral of a characteristic function of a measurable set A is given by

$$\int_A \chi_A(x) = m^*(A) \tag{3.3}$$

Definition 3.2.3 (Simple functions)**.** Let $\chi_A(x)$ be a characteristic function of a measurable set A. A simple function ψ is a linear combination of characteristic functions $\chi_{A_i}(x)$ of disjoint, measurable sets A_i that partition the set A, where $A_i = \{x \in A : \psi(x) = a_i\}$, i.e

$$\psi(x) = \sum_{i=1}^{n} a_i \chi_{A_i}(x) \tag{3.4}$$

Definition 3.2.4 (The Lebesgue Integral of a Simple Function)**.** For a simple function $\psi(x)$ its Lebesgue integral can be expressed as

$$\int_A \psi = \sum_{i=1}^{n} a_i m^*(A_i) \tag{3.5}$$

where we assume every set A_i is of finite measure.

Definition 3.2.5 (The Lebesgue Integral of a measurable Non-negative function)**.** The Lebesgue integral of a measurable non-negative function $f : A \to [0, \infty]$ is given by

$$\int_A f = \sup_{0 \leq \psi \leq f} \int_A \psi \tag{3.6}$$

for simple functions ψ.

Definition 3.2.6 (The Lebesgue Integral of arbitrary measurable functions)**.** A function $f : A \to [-\infty, \infty]$ is Lebesgue integrable over the measurable set A if

$$\int_A |f| < \infty \tag{3.7}$$

We can decompose a measurable, Lebesgue integrable function f into its positive and negative part as follows $f(x) = f^+(x) - f^-(x)$. f^+ and f^- and both non-negative. The Lebesgue integral of f over A is

$$\int_A f = \int_A f^+ - \int_A f^- \tag{3.8}$$

We saw in Section 2.2 that functions of bounded variation are bounded and measurable. So they can be Lebesgue integrated as per (3.8). Lebesgue and Riemann integration are also related in the following way.

Proposition 3.2.7. [SS05, Chapter 2, Theorem 1.5] *Every Riemann integrable function $f : [a, b] \to \mathbb{R}$ is measurable and Lebesgue integrable with*

$$R \int_a^b f(x)\ dx = \int_a^b f(x)\ dx$$

where $R \int_a^b f(x)\ dx$ is the Riemann integral of f.

Proof. Since f is Riemann integrable it is bounded $\implies \exists M > 0$ such that $|f(x)| \leq M\ \forall\ x \in [a, b]$. We can also construct two sequences of step functions[5] $\{\psi_k\}$ and $\{\phi_k\}$ that satisfy the following properties:

1. $|\psi_k(x)| \leq M$ and $|\phi_k(x)| \leq M\ \forall x \in [a, b]$ and $k \geq 1$ with

$$\phi_1(x) \leq \phi_2(x) \leq \ \ldots\ f \leq \ \ldots\ \leq \psi_2(x) \leq \psi_1(x)$$

2. $\lim_{k\to\infty} R \int_a^b \psi_k(x)\ dx = \lim_{k\to\infty} R \int_a^b \phi_k(x)\ dx = R \int_a^b f(x)\ dx$

[5] See [SS05, Appendix Section 1]

There are several things to note. Firstly, every step function is also a *single function* therefore

$$R\int_a^b \psi_k(x)\ dx = \int_a^b \psi_k(x)\ dx \text{ and } R\int_a^b \phi_k(x)\ dx\ = \int_a^b \phi_k(x)\ dx\ \forall k \geq 1$$

If we let $\tilde{\psi}(x) = \lim_{k\to\infty} \psi_k(x)$ and $\tilde{\phi}(x) = \lim_{k\to\infty} \phi_k(x)$ then we have $\tilde{\phi} \leq f \leq \tilde{\psi}$. Both $\tilde{\psi}$ and $\tilde{\phi}$ are measurable with

$$\lim_{k\to\infty} \int_a^b \psi_k(x)\ dx = \int_a^b \tilde{\psi}(x)\ dx$$

and

$$\lim_{k\to\infty} \int_a^b \phi_k(x)\ dx = \int_a^b \tilde{\phi}(x)\ dx.$$

Combining these results

$$\int_a^b (\tilde{\psi}(x) - \tilde{\phi}(x))\ dx = 0.$$

Knowing $\psi_k(x) - \phi_k(x) \geq 0$ we must also have $\tilde{\psi}(x) - \tilde{\phi}(x) \geq 0$

As a consequence of the bounded convergence theorem[6] if a function $g \geq 0$ is bounded and supported on a set of finite measure with $\int g = 0$ then $g = 0$ almost everywhere.

So $\tilde{\psi}(x) = \tilde{\phi}(x)$ almost everywhere $\implies$ f is measurable.

Since $\psi_k \to f$ almost everywhere,

$$\lim_{k\to\infty} \int_a^b \psi_k(x)\ dx = \int_a^b f(x)\ dx$$

and

$$R\int_a^b f(x)\ dx = \int_a^b f(x)\ dx.$$

□

Where the Riemann and Lebesgue integral differ is that Lebesgue integral is equipped to handle a.e. behaviour. We will illustrate this with an example.

[6] See [SS05, Section 2.1, Theorem 1.4]

Example 3.2.8 (The Dirichlet Function)**.** The Dirichlet function is defined as

$$D(x) = \begin{cases} 1 \text{ if } x \in \mathbb{Q} \cap [0,1] \\ 0 \text{ otherwise} \end{cases} \tag{3.9}$$

Notably, we saw before that the set $\mathbb{Q} \cap [0,1]$ has measure 0.

$$\implies D(x) = 0 \text{ a.e on } [0,1].$$

Taking the Lebesgue integral:

$$\int_0^1 D(x)\ dx = \int_0^1 0\ dx = 0.$$

Evidently the Lebesgue integral of a function $f(x)$ is indifferent to the value of $f(x)$ on a null set.

Lebesgue integration, Lebesgue measurable functions and functions of bounded variation are linked by a powerful result.

Theorem 3.2.9. *If f is of bounded variation on $[a,b] \subseteq \mathbb{R}$ then it is differentiable almost everywhere on $[a,b]$ and f' is Lebesgue integrable on $[a,b]$.*

Proof. If f is of bounded variation on $[a.b]$ we have seen that it can be split into two monotone increasing functions as per Theorem 2.2.8.

As seen in Proposition 3.1.15 monotone increasing functions are differentiable almost everywhere. So

$$f' = (f^+)' - (f^-)'$$

exists almost everywhere on $[a,b]$. The total variation of f^+ and f^- are finite and given by

$$V(f^+, [a,b]) = \int_a^b (f^+(x))'\ dx$$

and

$$V(f^-, [a,b]) = \int_a^b (f^-(x))'\ dx$$

respectively, therefore $f' = (f^+) - (f^-)'$ is Lebesgue integrable on $[a,b]$. □

There are a few other important results in Lebesgue integration theory which are needed to prove Theorem 4.2.2.

Lemma 3.2.10 (Fatou's Lemma)**.** [RF88, Theorem 4.9] *If for a sequence $< f_n >$ of non-negative, measurable functions $f_n(x) \to f(x)$ a.e on a set E then*

$$\int_E f \leq \lim_{n\to\infty} \int_E f_n.$$

Proof. We can assume without loss of generality convergence is everywhere, since the Lebesgue integral over a set of measure zero is zero. Let h be a bounded, measurable function such that $h \leq f$ and h vanishes outside of a set E' of finite measure. Let

$$h_n(x) = \min\{h(x), f_n(x)\}.$$

Then $h_n(x)$ is bounded by the bound on h and vanishes outside of E'. Now $h_n(x) \to h(x)\ \forall\ x \in E'$. By the Bounded Convergence Theorem [RF88, Chapter 4, Proposition 6]

$$\int_E h = \int_{E'} h = \lim_{n\to\infty} \int_{E'} h_n \leq \lim_{n\to\infty} \int_E f_n.$$

Taking the supremum over h,

$$\int_E f_n \leq \lim_{n\to\infty} \int_E f_n.$$

□

Theorem 3.2.11 (Monotone Convergence Theorem)**.** *Let $< f_n >$ be an increasing sequence of non-negative measurable functions and let $f = \lim_{n\to\infty} f_n$ a.e. Then*

$$\int f = \lim_{n\to\infty} \int f_n.$$

Proof. By Lemma 3.2.10, $\int f \leq \lim_{n\to\infty} \int f_n$. But for each n we have $f_n \leq f$ and so $\int f_n \leq \int f$. But this implies

$$\lim_{n\to\infty} \sup \int f_n \leq \int f.$$

Hence

$$\int f = \lim_{n\to\infty} \int f_n.$$

□

Proposition 3.2.12. [RF88, Proposition 4.14]

Let f be a non-negative function that is integrable over a set E. Then given $\varepsilon > 0\ \exists\ \delta > 0$ such that $\forall\ A \subset E$ with $m^*(A) < \delta$,

$$\int_A f < \varepsilon.$$

Proof. Let

$$f_n(x) = \begin{cases} f(x) \text{ if } f(x) \leq n \\ n \text{ otherwise.} \end{cases}$$

So each f_n is bounded and $f(x)$ converges to n at every point. By Theorem 3.2.11 $\exists$ N such that $\int_E f_N > \int_E f - \frac{\varepsilon}{2}$ and $\int_E f - f_N < \frac{\varepsilon}{2}$. Take $\delta = \frac{\varepsilon}{2N}$. If $m^*(A) < \delta$ then

$$\begin{aligned} \int_A f &= \int_A (f - f_N) + \int_A f_N \\ &< \int_E (f - f_N) + Nm^*(\mathrm{A}) \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

□

Theorem 3.2.13. [RF88, Theorem 5.3] *Let f be an increasing, real valued function on $[a, b]$. Then f is differentiable almost everywhere, f' is measurable and*

$$\int_a^b f'(x)\, dx \leq f(b) - f(a).$$

Proof. First we'll show that the sets where any two derivatives are unequal have measure zero. Recall from Definition 2.2.9 the Dini derivatives. We will consider the sets E where $D^+f(x) > D_-f(x)$. The sets arising from combinations of other Dini derivatives follow the same procedure as that of D^+ and D_-.

The set E is the union of sets ($\forall$ $u, v \in \mathbb{Q}$),

$$E_{u,v} = \{x : D^+f(x) > u > v > D_-f(x)\}.$$

We want to show that $m^*(E_{u,v}) = 0$. Let $s = m^*(E_{u,v})$. Take $\varepsilon > 0$ and enclose $E_{u,v}$ in an open set O with $m^*(O) < s + \varepsilon$. $\forall$ $x \in E_{u,v}$ there is an arbitrary small interval $[x - h, x]$ contained in O such that

$$f(x) - f(x - h) < vh.$$

By the Vitali Covering Lemma we can choose an arbitrary collection of these small intervals $\{I_1, ..., I_N\}$ whose interiors cover a subset $A \subset E_{u,v}$ such that the outer measure of A is greater than $s - \varepsilon$. Summing over these intervals

$$\begin{aligned}\sum_{n=1}^{N}[f(x_n)-f(x_n-h_n)] &< v\sum_{n=1}^{N}h_n\\ &< vm^*(O)\\ &< v(s+\varepsilon).\end{aligned}$$

Now each $y \in A$ is the left endpoint of an arbitrary, small interval $(y, y+k)$ that is contained in one of the I_n's for which $f(y+k)-f(y) > uk$. By Lemma 3.1.14 again we have a finite collection $\{J_1, ..., J_M\}$ of these intervals such that their union contains a subset of A of outer measure greater than $s-2\varepsilon$. Summing over thse intervals

$$\begin{aligned}\sum_{i=1}^{M} f(y_i+k_i)-f(y_i) &> u\sum_{i=1}^{M}k_i\\ &> u(s-2\varepsilon).\end{aligned}$$

Each J_i is contained in some I_n, summing over the the i's for which $J_i \subset I_n$ we get

$$\sum_{i=1}^{M} f(y_i+k_i)-f(y_i) \leq f(x_n)-f(x_n-h_n)$$

as f is increasing.

$$\implies \sum_{n=1}^{N} f(x_n)-f(x_n-h_n) \geq \sum_{i=1}^{M} f(y_i+k_i)-f(y_i).$$

So we have

$$v(s+\varepsilon) > u(s-2\varepsilon).$$

We know this to be true for $\forall\ \varepsilon > 0$. So $vs \geq us$. But $u > v, \implies s = 0$.

This shows that

$$g(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

is defined almost everywhere and f is differentiable when g is finite. Let $g_n(x) = n[f(x+\frac{1}{n})-f(x)]dx$ where we take $f(x) = f(b)$ for $x \geq b$. Then $g_n(x) \to g(x)$ for almost every $x \implies g$ is measurable. Also f is increasing so $g_n \geq 0$. By Lemma 3.2.10

$$\begin{aligned}\int_a^b f \leq \lim_{n\to\infty}\int_a^b g_n &= \lim_{n\to\infty} n\int_a^b [f(x+\frac{1}{n}) - f(x)]\, dx \\ &= \lim_{n\to\infty}[n\int_b^{b+\frac{1}{n}} f - n\int_a^{a+\frac{1}{n}} f\ ] \\ &= \lim_{n\to\infty}[f(b) - n\int_a^{a+\frac{1}{n}} f\ ] \\ &\leq f(b) - f(a).\end{aligned}$$

So g is integrable and hence finite almost everywhere. So f is differentiable a.e and $g = f'$ a.e. □

Chapter 4

Absolutely Continuous Functions

We have now developed an understanding of the background material required to examine absolute continuity and absolutely continuous functions. This chapter will demonstrate how they form the ideal framework in which to consider the Fundamental Theorem of Calculus. In Section 4.1 I will explore the concept of absolute continuity, exploring some properties and comparing absolutely continuous functions to other classes of functions we have seen before such as uniformly continuous functions, functions of bounded variation and Lipschitz functions. In Section 4.2 I will consider the characterisation of absolutely continuous functions by integrals as well as their differentiability, including integrability of the derivatives of absolutely continuous functions and preparing the discussion on a.e differentiability that comes in Section 5.1.

My primary references for this chapter were [RF88] and [SS05] while I also made use of [Ji20] and [Go94].

4.1 Definition and properties

Definition 4.1.1 (Absolute continuity)**.** f is absolutely continuous on an interval $I = [a, b]$ if $\forall\ \varepsilon > 0\ \exists\ \delta > 0$ such that

$$\sum_{k=1}^{N} |f(b_k) - f(a_k)| < \varepsilon \text{ when } \sum_{k=1}^{N} (b_k - a_k) < \delta \tag{4.1}$$

and the intervals $(a_k, b_k), k \in \{1, 2, ..., N\}$ are disjoint.

There are several important properties of absolutely continuous functions.

Proposition 4.1.2. *All absolutely continuous functions on I are uniformly continuous on I.*

Proof. Let f be absolutely continuous on I. By Definition 4.1.1, this means that for every $\varepsilon > 0$ there exists $\delta > 0$ such that for any finite collection of disjoint intervals $\{(a_k, b_k)\}_{k=1}^{N}$ in I,

$$\sum_{k=1}^{N}(b_k - a_k) < \delta \quad \Longrightarrow \quad \sum_{k=1}^{N}|f(b_k) - f(a_k)| < \varepsilon.$$

Now fix $\varepsilon > 0$ and let $\delta > 0$ be given by Definition 4.1.1. To prove that f is uniformly continuous, we must show that for all $x, u \in I$,

$$|x - u| < \delta \quad \Longrightarrow \quad |f(x) - f(u)| < \varepsilon.$$

So let $x, u \in I$ with $|x - u| < \delta$. Assume $x > u$ and consider the single interval (u, x). This is a finite collection of pairwise disjoint intervals with $N = 1$, so

$$\sum_{k=1}^{1}(b_k - a_k) = |x - u| < \delta.$$

By absolute continuity, we then have

$$\sum_{k=1}^{1}|f(b_k) - f(a_k)| = |f(x) - f(u)| < \varepsilon.$$

Since this holds for all $x, u \in I$ with $|x - u| < \delta$, it follows that f is uniformly continuous on I. □

Proposition 4.1.3. *All Lipschitz functions on I are absolutely continuous on I.*

Proof. Recall that f is Lipschitz on I if there exists a constant $L > 0$ such that

$$|f(x) - f(u)| \leq L|x - u| \quad \text{for all } x, u \in I.$$

We must show that f is absolutely continuous as seen in Definition 4.1.1.

Let $\varepsilon > 0$ be given and set $\delta = \frac{\varepsilon}{L}$. Consider any finite collection of disjoint intervals $\{(a_k, b_k)\}_{k=1}^{n}$ in I such that

$$\sum_{k=1}^{n}(b_k - a_k) < \delta.$$

Since f is Lipschitz for each k we have

$$|f(b_k) - f(a_k)| \leq L|b_k - a_k| = L(b_k - a_k)$$

since $b_k > a_k$.

$$\implies \sum_{k=1}^{n} |f(b_k) - f(a_k)| \leq L \sum_{k=1}^{n} (b_k - a_k) < L\delta = \varepsilon.$$

$\implies$ f is absolutely continuous on I. □

So uniform continuity, absolute continuity and Lipschitz are three classes of functions of increasing strictness with regards to the oscillation of the function. Uniform continuity governs the total oscillation over a single interval, absolute continuity governs the oscillation over a collection of disjoint intervals and Lipschitz over the entire function. From Proposition 4.1.2 and Proposition 4.1.3 the following implication chain is obvious

Figure 4.1: Implication chain

Lipschitz Continuity $\implies$ Absolute Continuity $\implies$ Uniform Continuity

We have already seen that uniform continuity includes functions for which the Fundamental Theorem of Calculus is not valid (for example the Cantor ternary function), and Lipschitz continuity is stronger than necessary for the Fundamental Theorem of Calculus (for example $f(x) = \sqrt{x}$ satisfies the Fundamental Theorem of Calculus but is not Lipschitz).

Absolutely continuous functions are also linked to another class of functions we have already seen, functions of bounded variation.

Theorem 4.1.4. ([Go94, Chapter 4, Theorem 4.3]) *If a function f is absolutely continuous on $I = [a, b]$, then f is of bounded variation on I.*

Proof. Choose $\varepsilon = 1$, then $\delta > 0$ is such that

$$\sum_{i=1}^{n} |f(b_k) - f(a_k)| < 1$$

when $\{[a_k, b_k : 1 \leq k \leq n]\}$ is any collection of disjoint intervals in I that satisfy $\sum_{i=1}^{n} |b_k - a_k| < \delta$. Choose $p \in \mathbb{Z},\ p > 0$ such that $\frac{b-a}{p} < \delta$.

Let $x_k = a + \frac{k(b-a)}{p}$ for $0 \leq k \leq p$. By Lemma 2.2.5

$$V(f,[a,b]) = \sum_{k=1}^{p} V(f,[x_{k-1},x_k]) \leq \sum_{k=1}^{p} 1 = p$$

Hence, f is of bounded variation on $[a,b]$.

□

Remark. Since all absolutely continuous functions are of bounded variation one can deduce that if f is absolutely continuous on I then it is of bounded variation on I. By Theorem 3.2.9 f' exists almost everywhere and f' is Lebesgue integrable on I.

Absolutely continuous functions have several other properties.

Proposition 4.1.5. [PW20] *If f, g are absolutely continuous functions on $[a,b]$, then $f+g$, $f-g$ and fg are absolutely continuous.*

Proof. Let $\varepsilon > 0$ be given. Since f, g are absolutely continuous by Definition 4.1.1 $\exists\ \delta,\ \delta'$ such that

$$\sum_{k=1}^{N} |f(b_k) - f(a_k)| < \frac{\varepsilon}{2} \text{ when } \sum_{k=1}^{N} (b_k - a_k) < \delta$$

and

$$\sum_{k=1}^{N} |g(b_k) - g(a_k)| < \frac{\varepsilon}{2} \text{ when } \sum_{k=1}^{N} (b_k - a_k) < \delta'.$$

for $\varepsilon > 0$. We take $\frac{\varepsilon}{2}$ instead of ε to make the following steps follow more easily, it changes nothing about absolute continuity. Let $\delta_{min} = \min\{\delta, \delta'\}$ to ensure we have a choice of δ such that both of the above statements hold simultaneously. Consider now the function $(f+g)(x)$,

$$\sum_{k=1}^{N} |(f+g)(b_k) - (f+g)(a_k)| \leq \sum_{k=1}^{N} |f(b_k) - f(a_k)| + \sum_{k=1}^{N} |g(b_k) - g(a_k)| < \varepsilon$$

$$\implies \sum_{k=1}^{N} |(f+g)(b_k) - (f+g)(a_k)| < \varepsilon.$$

Then $f+g$ is absolutely continuous. Similarly,

$$\sum_{k=1}^{N} |(f-g)(b_k) - (f-g)(a_k)| \leq \sum_{k=1}^{N} |f(b_k) - f(a_k)| + \sum_{k=1}^{N} |g(b_k) - g(a_k)| < \varepsilon$$

$$\implies \sum_{k=1}^{N} |(f-g)(b_k) - (f-g)(a_k)| < \varepsilon$$

So $f - g$ is absolutely continuous.

For fg consider that absolutely continuous functions are of bounded variation and hence bounded. So $\exists\ M, M' > 0$ such that

$$|f(x)| < M, |g(x)| < M'\ \forall\ x \in [a, b]$$

For $\varepsilon > 0, \delta$ and δ' defined as before (from Definition 4.1.1) using the boundedness of f, g we get

$$\sum_{k=1}^{N} |f(b_k)||g(b_k) - g(a_k)| \leq M \sum_{k=1}^{N} |g(b_k) - g(a_k)| < \frac{\varepsilon}{2}$$

and

$$\sum_{k=1}^{N} |g(a_k)||f(b_k) - f(a_k)| \leq M' \sum_{k=1}^{N} |f(b_k) - f(a_k)| < \frac{\varepsilon}{2}$$

So

$$\sum_{k=1}^{N} |f(b_k) - f(a_k)| < \frac{\varepsilon}{2M'}$$

and

$$\sum_{k=1}^{N} |g(b_k) - g(a_k)| < \frac{\varepsilon}{2M}$$

As before let $\delta_{min} = \min\{\delta, \delta'\}$. Then consider fg,

$$\begin{aligned} |(fg)(b_k) - (fg)(a_k)| &\leq |g(a_k)||f(b_k) - f(a_k)| + |f(b_k)||g(b_k) - g(a_k)| \\ &\leq M'|f(b_k) - f(a_k)| + M|g(b_k) - g(a_k)| \end{aligned}$$

$$\implies \sum_{k=1}^{N} |(fg)(b_k) - (fg)(a_k)| \leq M' \sum_{k=1}^{N} |f(b_k) - f(a_k)| + M \sum_{k=1}^{N} |g(b_k) - g(a_k)| < \varepsilon$$

So fg is absolutely continuous.

□

Proposition 4.1.6. *If f, g are absolutely continuous on $[a, b]$ and $\exists\ C > 0$ such that $|g(x)| \geq C$ for all $x \in [a, b]$, then $\frac{f}{g}$ is absolutely continuous on $[a, b]$.*

Proof. Let $h(x) = \frac{f(x)}{g(x)}$. As we saw in the proof of Proposition 4.1.5 since f is absolutely continuous it is bounded. i.e $|f(x)| \leq M\ \forall\ x \in [a, b]$. Also $|g(x)| \geq C\ \forall\ x \in [a, b] \implies \frac{1}{|g(x)|} \leq \frac{1}{C}$. Let $a_k, b_k \in [a, b]$ make up disjoint intervals as described in Definition 4.1.1 with $k \in \{1, 2, ..., N\}$, then take

$$\begin{aligned} h(b_k) - h(a_k) = \frac{f(b_k)}{g(b_k)} - \frac{f(a_k)}{g(a_k)} &= \frac{f(b_k)g(a_k) - f(a_k)g(b_k)}{g(b_k)g(a_k)} \\ &= \frac{g(a_k)(f(b_k) - f(a_k)) + f(a_k)(g(a_k) - g(b_k))}{g(b_k)g(a_k)} \end{aligned}$$

So

$$\begin{aligned} |h(b_k) - h(a_k)| &\leq \frac{|f(b_k) - f(a_k)|}{|g(b_k)|} + \frac{|f(a_k)||g(a_k) - g(b_k)|}{|g(b_k)g(a_k)|} \\ &\leq \frac{1}{C}|f(b_k) - f(a_k)| + \frac{M}{C^2}|g(a_k) - g(b_k)| \end{aligned}$$

$$\implies \sum_{k=1}^{N} |h(b_k) - h(a_k)| \leq \frac{1}{C} \sum_{k=1}^{N} |f(b_k) - f(a_k)| + \frac{M}{C^2} \sum_{k=1}^{N} |g(a_k) - g(b_k)|$$

Since f, g are absolutely continuous $\exists\ \delta, \delta'$ such that for

$$\sum_{k=1}^{N} (b_k - a_k) < \delta \implies \sum_{k=1}^{N} |f(b_k) - f(a_k)| < \frac{C\varepsilon}{2}$$

and

$$\sum_{k=1}^{N} (b_k - a_k) < \delta' \implies \sum_{k=1}^{N} |g(b_k) - g(a_k)| < \frac{C^2\varepsilon}{2M}$$

Take $\delta_{min} = \min\{\delta, \delta'\}$ and then evaluate the absolute continuity of h,

$$\sum_{k=1}^{N} |h(b_k) - h(a_k)| = \sum_{k=1}^{N} \left| \frac{f(b_k)}{g(b_k)} - \frac{f(a_k)}{g(a_k)} \right| < \frac{1}{C} \cdot \frac{C\varepsilon}{2} + \frac{M}{C^2} \cdot \frac{C^2\varepsilon}{2M} = \varepsilon$$

So $h(x) = \frac{f(x)}{g(x)}$ is absolutely continuous on $[a, b]$. □

We can also show the following:

Proposition 4.1.7. [Go94, Chapter 4, Theorem 4.16] *If $f : [a, b] \to \mathbb{R}$ is absolutely continuous on $[a, b]$ then f maps sets of measure zero to sets of measure zero.*

Proof. Suppose $A \subseteq (a, b)$ has measure zero. Let $\varepsilon > 0$ and choose $\delta > 0$ such that f is absolutely continuous. Let O be an open set in (a, b) such that $A \subseteq O$ and $m^*(O) < \delta$. Let $O = \bigcup_{k=1}^{\infty}(a_k, b_k)$ be a union of disjoint intervals. The Extreme Value theorem states that $\forall\ k\ \exists\ [c_k, d_k] \subseteq [a_k, b_k]$ such that

$$m^*(f([a_k, b_k])) = |f(d_k) - f(d_k)|$$

Since $\sum_{k=1}^{n}(d_k - c_k) \leq m^*(O) < \delta\ \forall\ n$ and $\sum_{k=1}^{n} |f(d_k) - f(c_k)| < \varepsilon$ we have

$$m^*(f(A)) \leq m^* f(O) \leq \sum_{k=1}^{\infty} m^* f([a_k, b_k]) = \sum_{k=1}^{\infty} |f(d_k) - f(c_k)| \leq \varepsilon$$

Since ε was arbitrary $m^* f(A) = 0$. So absolutely continuous functions map sets of measure zero to sets of measure zero. □

Remark. This result is not generally true for functions of bounded variation, which do not behave as "nicely" on null sets.

4.2 Differentiability and characterisation of absolutely continuous functions by integrals

We have seen before (through Theorem 4.1.4) that absolutely continuous functions f have derivatives f' that exist almost everywhere. In fact, these derivatives f' are Lebesgue integrable. This is an important fact that we can use to characterise absolutely continuous functions by integrals.

Theorem 4.2.1 (Characterisation of absolute continuity by integrals)**.** [Ji20, Theorem 2.3]

A function F is absolutely continuous on $[a, b]$ if and only if

$$F(x) = F(a) + \int_a^x f(t)\ dt \tag{4.2}$$

for some integrable f such that $f(t) = F'(t)$.

Before launching into the proof of this Theorem 4.2.1 it is necessary to provide a number of results.

Theorem 4.2.2 (Integral representation of absolutely continuous functions)**.** [RF88, Theorem 5.14][PW20, p. 1]

For a function $g(x)$ that is Lebesgue integrable on $[a, b]$ the function

$$G(x) = \int_a^x g(t)\ dt \tag{4.3}$$

is absolutely continuous on $[a, b]$.

Proof. Let $\varepsilon > 0$ be given. By Proposition 3.2.12 $\exists\ \delta > 0$ such that for any measurable set $A \subset [a, b]$ with $m^*(A) < \delta$,

$$\int_A |g(t)|\ dt < \varepsilon.$$

Let $\{(a_k, b_k)\}_{k=1}^N$ be a collection of mutually disjoint subintervals of $[a, b]$ such that

$$m^*(\bigcup_{k=1}^N (a_k, b_k)) = \sum_{k=1}^N (b_k - a_k) < \delta.$$

For each interval $G(b_k) - G(a_k) = \int_{a_k}^{b_k} g(t)\ dt$.

Thus

$$\begin{aligned}
\sum_{k=1}^N |G(b_k) - G(a_k)| &= \sum_{k=1}^N |\int_{a_k}^{b_k} g(t)\ dt| \\
&\leq \sum_{k=1}^N \int_{a_k}^{b_k} |g(t)|\ dt \\
&= \int_{\bigcup_{k=1}^N (a_k, b_k)} |g(t)|\ dt \\
&< \varepsilon.
\end{aligned}$$

□

Theorem 4.2.3. [SS05, Chapter 3, Theorem 3.8] *If f is absolutely continuous on $[a,b]$ and $f'(x) = 0$ for almost every $x \in [a,b]$, then f is constant.*

Proof. The goal is to show that $f(a) = f(c)\ \forall\ c \in [a,b]$. Let $A = \{x \in [a,c] : f'(x) = 0\}$. For $\varepsilon > 0$ since f is absolutely continuous then $\exists\ \delta > 0$ as defined in Definition 4.1.1.

$\forall\ x \in A$ we have $f'(x) = 0$, hence $\exists$ an arbitrary small interval $[a_x, c_x]$ such that $x \in [a_x, c_x] \subseteq [a,c]$ and

$$|f(c_x) - f(a_x)| < \varepsilon(c_x - a_x)$$

We can see that the interval $[a_x, c_x]$ is a Vitali cover of A. So, by the Vitali covering lemma there is a finite collection of mutually disjoint intervals $\{I_k : 1 \leq k \leq n\}$ in $[c_x - a_x]$ such that

$$m^*(A - \bigcup_{k=1}^{n} I_k) < \delta$$

Since $m^*([a,c] \backslash A) = 0$

$$m^*([a,c] - \bigcup_{k=1}^{n} I_k) = m^*(A - \bigcup_{k=1}^{n} I_k) < \delta$$

Take $a \leq x_1 \leq y_1 \leq x_2 \leq \ \ldots\ \leq y_n = c$. Let $y_0 = a$ and $x_{n+1} = c$. Then

$$\sum_{k=0}^{n} (x_{k+1} - y_k) = m^*([a,c] - \bigcup_{k=1}^{n} I_k) < \delta$$

Therefore,

$$\sum_{k=0}^{n} |f(x_{k+1}) - f(x_k)| < \sum_{k=1}^{n} \varepsilon(y_k - x_k) \leq \varepsilon(c-a)$$

From these inequalities

$$|f(c) - f(a)| \leq \sum_{k=0}^{n} |f(x_{k+1}) - f(y_k)| + \sum_{k=1}^{n} |f(y_k) - f(x_k)| < \varepsilon(c - a + 1)$$

So $|f(c) - f(a)| \leq \varepsilon(c-a+1)\ \forall\ \varepsilon > 0. \implies f(c) = f(a)$

□

Theorem 4.2.4. [RF88, Section 5.3, Theorem 10] *If f is Lebesgue integrable on $[a,b]$ and F is defined as*

$$F(x) = \int_a^x f(t)\ dt$$

then F is absolutely continuous on $[a,b]$ and $F'(x) = f(x)$ for almost every $x \in [a,b]$.

Proof. By Theorem 4.2.2, F is absolutely continuous on $[a,b]$. By Theorem 4.1.4 F is of bounded variation and hence continuous.

For $n \in \mathbb{N}$, let

$$g_n(x) = \frac{F(x+\frac{1}{n}) - F(x)}{\frac{1}{n}} = n\int_x^{x+\frac{1}{n}} f(t)\,dt \quad \text{for } x \in [a, b-\tfrac{1}{n}]$$

By the Lebesgue differentiation theorem [Fo99, Theorem 3.21], for a.e $x \in [a,b]$,

$$\lim_{n\to\infty} g_n(x) = f(x)$$

Hence, for a.e x,

$$F'(x) = \lim_{h\to 0} \frac{F(x+h) - F(x)}{h} = \lim_{n\to\infty} g_n(x) = f(x)$$

We can assume without loss of generality that $f \geq 0$. $\forall\ n \in \mathbb{N}$, define

$$f_n(x) = \min\{f(x), n\}$$

Let

$$F_n(x) = \int_a^x f_n(t)\,dt, \text{ and } G_n(x) = \int_a^x [f(t) - f_n(t)]\ dt$$

such that $F = F_n + G_n$. Since $f - f_n \geq 0$, G_n is monotone increasing and by Proposition 3.1.15 is differentiable almost everywhere. Also, $F_n'(x) = f_n(x)$ almost everywhere. Therefore, for almost every x,

$$F'(x) = F_n'(x) + G_n'(x) \geq f_n(x).$$

As $n \to \infty$, $f_n(x) \to f(x)$ and therefore $F'(x) \geq f(x)$ almost everywhere. Integrating,

$$\int_a^b F'(x)\,dx \geq \int_a^b f(x)\,dx = F(b) - F(a).$$

By absolute continuity of F,

$$\int_a^b F'(x)\,dx \leq F(b) - F(a).$$

Combining these, we get

$$\int_a^b [F'(x) - f(x)]\,dx = 0.$$

Since $F'(x) - f(x) \geq 0$ almost everywhere,

$$F'(x) = f(x) \quad \text{for a.e } x \in [a,b].$$

□

Theorem 4.2.5. *If f is absolutely continuous on $[a,b]$ then f is differentiable a.e on $[a,b]$.*

Proof. By Theorem 4.1.4 if f is absolutely continuous on $[a,b]$ it is of bounded variation on $[a,b]$. Then by Theorem 3.2.9 f is differentiable almost everywhere on $[a,b]$ and f' is Lebesgue integrable on $[a,b]$. □

We can now prove Theorem 4.2.1.

Proof. Let f be some integrable function on $[a,b]$ such that

$$F(x) = F(a) + \int_a^x f(t)\ dt\ \forall\ x \in [a,b].$$

Let $\varepsilon > 0$. By Proposition 3.2.12 $\exists\ \delta > 0$ such that for $A \subset [a,b]$ with $m^*(A) < \delta$.

$$\int_A |f(t)|\ dt < \varepsilon$$

Let $\{(a_k, b_k)\}_{k=1}^N$ be a collection of disjoint subintervals of $[a,b]$ with $\sum_{k=1}^N (b_k - a_k) < \delta$. Then for each $k \in [1,N]$

$$|F(b_k) - F(a_k)| = \int_{a_k}^{b_k} |f(t)|\ dt.$$

Summing over k

$$\sum_{k=1}^{N} |F(b_k) - F(a_k)| = \sum_{k=1}^{N} \int_{a_k}^{b_k} |f(t)| \, dt = \int_{\bigcup_{k=1}^{N}(a_k,b_k)} |f(t)| \, dt < \varepsilon.$$

So by Definition 4.1.1 F is absolutely continuous.

Now assume that F is absolutely continuous on $[a,b]$. By Theorem 4.2.5 F is differentiable a.e on $[a,b]$ with integrable F'. We just saw that since F is absolutely continuous it can be expressed as $F(x) = F(a) + \int_a^x f(t) \, dt$. Let

$$G(x) = F(a) + \int_a^x F'(t) \, dt, \ x \in [a,b].$$

By Theorem 4.2.4 $G'(x) = F'(x)$ for a.e $x \in [a,b]$. $\implies$ $(F-G)' = 0$ for a.e $x \in [a,b]$. By Theorem 4.2.3 $F - G$ is constant. By choosing $x = a$ in the expression for $G(x)$ it is easily shown that $F(a) = G(a)$. So $F(x) = G(x) \ \forall \ x \in [a,b]$. So a function F is absolutely continuous if it can be expressed as $F(x) = F(a) + \int_a^x f(t) \, dt$.

□

Now that we have an understanding of absolutely continuous functions there is an important question to ask. Why is absolute continuity the ideal condition for the Fundamental Theorem of Calculus? This is what the next chapter will demonstrate.

Chapter 5

Fundamental Theorem of Calculus

This chapter will focus on knitting together all that we've seen so far. With a few final results we will then be able to consider the Fundamental Theorem of Calculus for absolutely continuous functions and why absolute continuity is the ideal framework in which to consider this theorem.

Section 5.1 will explore in depth a topic we've seen before, almost everywhere differentiability. I will explain why we cannot guarantee differentiability of absolutely continuous functions everywhere and demonstrate that in fact almost everywhere differentiability is the best possible result. In Chapter 1 we asked why the (Riemann) Fundamental Theorem of Calculus is insufficient, almost everywhere differentiability is crucial in answering this question. I will also give a brief remark on how the interaction between Lebesgue integration and differentiation is an important step in refining the Fundamental Theorem of Calculus. Section 5.2 will be the culmination of all we have seen. I will state the Fundamental Theorem of Calculus for absolutely continuous functions, provide its proof and explain why it is the most general context in which to consider the Fundamental Theorem of Calculus by comparing it to the (Riemann) Fundamental Theorem.

My primary references for this chapter are [SS05], [Fo99] and [BS11]. I also used [Bo19], [Go94] and [Ji20].

5.1 Almost everywhere differentiability

Recall Definition 3.1.1, that of a null set. We also had Definition 3.1.3 which demonstrated that a statement $Q(x)$ about a point $x \in [a,b]$ holds almost everywhere on $[a,b]$ if it holds everywhere in $[a,b]$ except on a null set. Clearly then, a function f is differentiable almost everywhere on $[a,b]$ if it is differentiable everywhere on $[a,b]$ except on some set $Z \subset [a,b]$ of measure zero.

At the heart of why almost everywhere differentiability is required we need the following result.

Proposition 5.1.1. *Absolutely continuous functions may fail to be differentiable on null sets.*

Proof. We saw in Section 4.1 that absolutely continuous functions are of bounded variation, thus their derivatives exist almost everywhere. Then by Definition 3.1.3 it is implicit that an absolutely continuous function f may fail to be differentiable on a null set. □

In general, derivatives may fail to exist for highly oscillatory functions or at discontinuous or singular points. Differentiability almost everywhere aids in excluding such points.

Recall the Cantor ternary function. We saw that it is differentiable almost everywhere with derivative 0, which is integrable.

But yet the Fundamental Theorem of Calculus did not hold. So a function being differentiable almost everywhere and having integrable derivative is not enough to generalise the fundamental theorem of calculus. For that we need absolute continuity.

Recall Theorem 4.2.2. An equivalent statement using Riemann integration goes as follows [BS11, p. 210].

Proposition 5.1.2. *If a function f is continuous on $[a, b]$ then you can define*

$$F(x) = R\int_a^x f(t)\ dt\ \forall\ x \in (a, b).$$

This is implicit from Theorem 1.1.8.

The classical statement may fail for highly discontinuous functions. So by taking the Lebesgue integral instead of the Riemann integral the fundamental theorem of calculus can be generalised to include many more functions, at the necessary cost of weakening the derivative from existing $\forall\ x \in [a, b]$ to existing a.e. in $[a, b]$. Classically we could only assume that our F as defined above is differentiable but now we can assume it is absolutely continuous which is a stronger global property.

From the definition of absolute continuity, namely Theorem 4.2.4 we know that $F' = f$ almost everywhere. To deal with this almost everywhere behaviour we need the Lebesgue integral, as the Riemann integral is not equipped to handle such behaviour.

So since we can define F as a Lebesgue integral of f, and $F' = f$ almost everywhere, Lebesgue integration is essential in generalising the inverse relationship between differentiation and integration and absolutely continuous functions are exactly the class of functions to do so.

5.2 The Fundamental Theorem of Calculus for Absolutely Continuous Functions

Finally we are in a position to state the Fundamental Theorem of Calculus for absolutely continuous functions.

Theorem 5.2.1 (The Fundamental Theorem of Calculus for Absolutely Continuous Functions (First Form))**.** *If a function f is Lebesgue integrable on $[a,b]$ then $F(x) = \int_a^x f(t)\,dt$ is absolutely continuous on $[a,b]$ and F' exists almost everywhere on $[a,b]$ with $F' = f$.*

This theorem is a direct consequence of Theorem 4.2.2 which shows that F is absolutely continuous and Theorem 4.2.4 shows that $F' = f$ a.e.

Recall Theorem 1.1.8, the (Riemann) First Form of the Fundamental Theorem of Calculus: if g is continuous and Riemann integrable on $[a,b]$ we can define a continuous $G(x) = R\int_a^x g(t)\,dt$, then $G'(x) = g(x)$ everywhere on $[a,b]$ where $R\int$ defines the Riemann integral.

Theorem 5.2.2 (The Fundamental Theorem of Calculus for Absolutely Continuous Functions (Second Form))**.** *If f is absolutely continuous on $[a,b]$ then by* Theorem 4.2.5 *f' exists almost everywhere and is Lebesgue integrable on $[a,b]$. Then*

$$\int_a^x f'(t)\,dt = f(x) - f(a) \tag{5.1}$$

for $x \in [a,b]$.

Proof. We saw in Theorem 2.2.8 that if a function of bounded variation can be split into two monotone increasing functions. Since f is absolutely continuous it is of boundary variation and can then be split as such. By Theorem 3.2.9 f' is Lebesgue integrable on $[a,b]$. Let

$$g(x) = \int_a^x f'(t)\,dt.$$

By Theorem 4.2.2 g is absolutely continuous, and by Proposition 4.1.5 so is $g - f$.

By the Lebesgue Differentiation Theorem [SS05, Section 2.1, Theorem 1.4] $g'(x) = f'(x)$ for almost every $x \in [a,b]$.

Hence $(f-g)' = 0$ almost everywhere. By Theorem 4.2.3 $f - g$ is constant. So take

$$f(x) - \int_a^x f'(t)\,dt = c$$

for an arbitrary constant c.

At $x = a$:

$$g(a) = \int_a^a f'(t)\,dt = 0$$

$$\implies f(a) - g(a) = f(a) = c.$$

But $f - g$ is a constant.

$$\implies f(x) - g(x) = c = f(a).$$

Rearranging this we get,

$$g(x) = \int_a^x f'(t)\ dt = f(x) - f(a).$$

This is the Fundamental Theorem of Calculus for Absolutely Continuous Functions (Second Form). □

So we have shown that absolute continuity provides the class of functions for which differentiation and integration are the inverse of one another, and it is a sufficient condition to generalise the Fundamental Theorem of Calculus.

We need no stronger (see Figure 4.1) assumptions (for example Lipschitz), absolute continuity is such that the fundamental theorem of calculus holds for all absolutely continuous functions and absolute continuity does not exclude functions for which the fundamental theorem of calculus holds. We have also recovered the inverse relationship between differentiation and integration totally resolving the question we asked in Section 1.2.

Chapter 6

Consequences and Applications

Having seen now the Fundamental Theorem of Calculus for Absolutely Continuous Functions I will give a short insight into its potential applications. In Section 6.1 I will show that it can be used to solve ordinary differential equations that we couldn't use the Riemann Fundamental Theorem of Calculus to solve. In Section 6.2 I will show that integration by parts for absolutely continuous functions is a direct consequence of the Fundamental Theorem of Calculus. Section 6.3 will include a brief introduction on Opial Inequalities, which are a direct consequence of absolute continuity and the Fundamental Theorem of Calculus.

My primary reference for Section 6.1 and Section 6.2 was [Fo99]. In Section 6.3 I used [Ag99] and [Le99].

6.1 Differential Equations

As stated, the Fundamental Theorem of Calculus can be used to solve ordinary differential equations that are inaccessable through the Riemann Fundamental Theorem of Calculus.

To solve $y' = f(x)$ for integrable f let

$$y(x) = y_0 + \int_a^x f(t)\, dt.$$

By Theorem 4.2.1 y is absolutely continuous and $y' = f$ almost everywhere.

So we can find a well defined solution to $y' = f$.

I will illustrate this through a series of examples.

Example 6.1.1. Recall the Dirichlet function. For $x \in [0,1]$ let

$$y'(x) = D(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases} \quad \text{a.e with } y(0) = 0$$

This could not be solved using the Riemann Fundamental Theorem of Calculus as $D(x)$ is not Riemann integrable. However, $D(x)$ is Lebesgue integrable.

$$\implies y(x) = y_0 + \int_a^x D(t)\, dt = y_0 = 0 \text{ for almost every } x \in [0,1].$$

This problem is what is known as a (classically) *ill-posed problem* which we have solved with the Fundamental Theorem of Calculus for Absolutely Continuous Functions.

Remark. Note that $D(x)$, as well as $T(x)$ (see Example 6.1.2) and the examples that follow are not themselves absolutely continuous, but when we solve for y we get a function $y(x)$ that is absolutely continuous.

Example 6.1.2. Consider the *Thomae Function* ODE: $y'(x) = T(x),\ y_0 = 0$ for a.e $x \in [0,1]$ with

$$T(x) = \begin{cases} x = \frac{p}{q}, x \in \mathbb{Q} \text{ with } x = \frac{p}{q} \text{ in simplest form} \\ x = 0, x \notin \mathbb{Q} \end{cases}$$

$T(x)$ is discontinuous and therefore not Riemann integrable. Using the Fundamental Theorem of Calculus for Absolutely Continuous Functions we get

$$y(x) = y_0 + \int_0^x T(t)\, dt = y_0 + 0 = 0 \text{ for a.e } x \in [0,1].$$

Example 6.1.3. Let

$$y'(x) = f(x) = \begin{cases} \sin(\frac{1}{x^2}), & x \neq 0 \\ 0, & x = 0 \end{cases} \quad \text{for a.e } x \in [0,1]$$

Here $f(x)$ is Riemann integrable but $f'(x)$ does not exist at $x = 0$. So we are considering a function that is differentiable a.e and we need the Fundamental Theorem of Calculus for Absolutely Continuous Functions.

$$\implies y(x) = \int_0^x \sin(\frac{1}{t^2})\, dt.$$

6.2 Integration by Parts

Proposition 6.2.1 (Integration by parts for absolutely continuous functions)**.** *If f, g are absolutely continuous on $[a, b]$ then*

$$\int_a^b gf' \, dx = fg\Big|_a^b - \int_a^b fg' \, dx \tag{6.1}$$

Proof. We saw before that if f, g are absolutely continuous then so is fg. So it is differentiable almost everywhere and

$$\implies (fg)' = f'g + g'f \text{ almost everywhere.}$$

Using the Fundamental Theorem of Calculus

$$\int_a^b (fg)' \, dx = fg\Big|_a^b$$

$$\implies \int_a^b f'g + g'f \, dx = fg\Big|_a^b$$

$$\implies \int_a^b gf' \, dx = fg\Big|_a^b - \int_a^b g'f \, dx.$$

Which is exactly the formula for integration by parts. □

6.3 Opial Inequalities

Opial Inequalities are a topic into which there has been a large amount of research and could easily form a text such as this of their own, so I will keep my discussion on them brief. [Ag99] is an exceptional resource to discover more on the topic.

Theorem 6.3.1 (Opial Inequality (1960))**.** [Le99, Theorem B] *Let $x(t)$ be absolutely continuous in $[0, h]$ and $x(0) = x(h) = 0$. Then*

$$\int_0^h |x(t)x'(t)|\ dt \leq \frac{h}{4}\int_0^h (x'(t))^2\ dt \tag{6.2}$$

Equality holds in Equation (6.2) for a function of the following type

$$x_0(t) = \begin{cases} ct,\ t \in [0, \frac{h}{2}] \\ c(h-t),\ t \in [\frac{h}{2}, h] \end{cases}$$

for an arbitrary constant $c > 0$. This function is not differentiable at $t = \frac{h}{2}$. Recall though an important property of absolutely continuous functions: differentiability *a.e.*

Proof (Olech's proof). I will now present Olech's proof of the above Opial Inequality, which relies heavily on absolute continuity.

Let $y(t) = \int_0^t |x'(s)|\ ds$ and $z(t) = \int_t^h |x'(s)|\ ds$.

$$\implies y'(t) = |x'(t)| = -z'(t)$$

and

$$|x(t)| \leq y(t),\ |x(t)| \leq z(t),\ t \in [0, h].$$

Using these, we have

$$\int_0^{\frac{h}{2}} |x(t)x'(t)|\ dt \leq \int_0^{\frac{h}{2}} y(t)y'(t)\ dt = \frac{1}{2}y^2(\frac{h}{2})$$

and

$$\int_{\frac{h}{2}}^h |x(t)x'(t)|\ dt \leq -\int_{\frac{h}{2}}^h z(t)z'(t)\ dt = \frac{1}{2}z^2(\frac{h}{2}).$$

Adding these gives

$$\int_0^h |x(t)x'(t)|\ dt \leq \frac{1}{2}(y^2(\frac{h}{2}) + z^2(\frac{h}{2})).$$

We will now use the Cauchy-Schwarz Lemma[1]: For Lebesgue integrable f, g:

$$|\int_a^b fg\ dx\ | \leq (\int_a^b |f|^2\ dx)^{\frac{1}{2}}(\int_a^b |g|^2\ dx)^{\frac{1}{2}}.$$

[1] See [Ag99, p. 7]

So

$$y^2(\frac{h}{2}) = (\int_0^{\frac{h}{2}} |x'(t)|\ dt)^2 \leq \frac{h}{2}\int_0^{\frac{h}{2}} (x'(t))^2\ dt$$

and

$$z^2(\frac{h}{2}) \leq \frac{h}{2}\int_{\frac{h}{2}}^{h} (x'(t))^2\ dt.$$

Using these results (6.2) follows. In the case that equality in (6.2) holds we have

$$(\int_0^{\frac{h}{2}} |x'(t)|\ dt\ )^2 = \frac{h}{2}\int_0^{\frac{h}{2}} (x'(t))^2\ dt$$

and

$$(\int_{\frac{h}{2}}^{h} |x'(t)|\ dt\ )^2 = \frac{h}{2}\int_{\frac{h}{2}}^{h} (x'(t))^2\ dt.$$

These are only possible if $|x'(t)|$ is constant almost everywhere on $[0, \frac{h}{2}]$ and $[\frac{h}{2}, h]$. This implies that $x(t) = x_0(t)$. □

As previously stated Opial Inequalities are a massive field in and of themselves, so for this project I will park the discussion having showed that Opial's original inequality can be viewed as a consequence of absolute continuity.

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